

M. M. Institute of Computer Technology & Business Management (MCA), M.M. (Deemed to Be University), Mullana		Course coverage Plan (Practical)
Session: 2025-26 (Even)	Class/Semester: BCA – 6th Semester	
Subject Code: BCA-602	Subject Name: R Programming Lab	
Lecture: 2	Tutorial: 0	
Max. Marks Internal :60	Max. Marks External: 40	Credits: 1.0

Lect No.	G1	G2	G3	Practical Assignment
1.	09/01	08/01	08/01	Installation and Setup of R/RStudio – Install R/RStudio, explore RStudio interface, run basic commands. <i>Write a program to display “Welcome to R Programming”.</i>
2.	16/01	15/01	15/01	<i>Write an R script to calculate the sum, difference, product, and division of two numbers.</i>
3.	16/01	15/01	15/01	Variables and Operators in R – Perform arithmetic, relational, and logical operations using variables. Experiment: <i>Create variables of different data types in R and display their type using appropriate functions</i>
4.	23/01	22/01	22/01	Vectors and Vectorized Operations – Create vectors and apply vectorized arithmetic operations. Experiment: 4. Write a R program to perform following operations on Vector: min(), max(), mean(), sqrt(), length(), sum(), prod(), sort()(in ascending and descending order), rev(), addition/ subtraction/multiplication/division of two vectors. 5. Write a R program to create a list containing a vector, a matrix and a list and perform following operations: iv. Update the elements in the list. v. Merge two lists into one list. vi. Count number of objects in a given list.
5.	23/01	22/01	22/01	Subsetting Data in R – Subset vectors using indexing, logical conditions, and named elements. Experiment: <i>Create a vector of 10 elements and extract even-indexed elements.</i>
6.	30/01	29/01	29/01	Handling NA and NULL Values – Detect, remove, and replace NA and NULL values; apply coding standards. Experiment: <i>Create a vector with NA values and write a program to remove them.</i>
7.	06/02	05/02	05/02	Conditional Statements Using if-else and Nested if-else Statements – Implement decision-making using if statements. Experiment: <i>Write a program to check leap year in R.</i>
8.	13/02	12/02	12/02	For Loop Implementation and While Loop Implementation – Use for loop for repetitive tasks and sequence generation. Experiment: 1. Write a user defined function to generate Fibonacci series using while loop 2. Generate the multiplication table of a given number using a for loop . 3. Write a program in R to make a simple calculator

9.	20/02	19/02	19/02	Scoping Rules in R – Demonstrate local and global variables using functions. Experiment: Write an R program to demonstrate the difference between local and global variables by defining a variable outside a function and modifying it inside the function. Display the values of the variable before and after function execution.
10.	20/02	19/02	19/02	Apply Family Functions – Use apply(), lapply(), sapply(), and tapply(). Experiment: Use apply(), lapply(), and sapply() on a matrix and compare the results.
11.	27/02	26/02	26/02	Working with Date, Time, and Sleep Functions – Handle system time, dates, and implement delays using Sys.sleep(). Experiment: Write a program to display the current system date and time and pause execution for 5 seconds.
12.	27/02	26/02	26/02	Reading and Writing Data Files – Read/write CSV and text files using read.csv() and write.csv(). Experiment: Read a CSV file into R and display summary statistics of the data.
13.	06/03	05/03	05/03	Saving and Loading R Data Objects – Use save(), load(), and RData files. Experiment: Create an R object, save it to a file, and reload it.
14.	06/03	05/03	05/03	Generating Data in R – Generate sequences and random data using built-in functions. Experiment: Generate 100 random numbers between 1 and 50 and store them in a vector.
15.	13/03	12/03	12/03	Regular and Random Sequences – Use seq(), rep(), runif(), rnorm(). Experiment: Generate a regular sequence from 1 to 100 with a step size of 5.
16.	13/03	12/03	12/03	Handling Missing Data – Use is.na(), na.omit(), and complete.cases(). Experiment: Write a program to identify and replace missing values with the mean.
17.	20/03	19/03	19/03	Creating and Manipulating Objects – Work with vectors, matrices, lists, and data frames. Experiment: Write a R program to create a data frame and get the structure, statistical summary and nature of the data of a given data frame.
18.	20/03	19/03	19/03	Indexing and Accessing Object Values – Demonstrate indexing systems for different R objects. Experiment: Write a R program to create a matrix and perform following operations: iii. Extract the sub matrix whose rows have column value > 3 from a given matrix. Convert a matrix to a 1 dimensional array.
19.	03/04	02/04	02/04	Introduction to S3 and S4 Classes – Create and use simple S3 and S4 objects. Experiment: Demonstrate indexing on vectors, matrices, and data frames.
20.	03/04	02/04	02/04	Reference Classes and User-Defined Functions – Implement reference classes and arithmetic functions. Experiment: Create a reference class to perform basic arithmetic operations.
21.	10/04	09/04	09/04	Creating Basic 2D and 3D Plots – Plot line graphs, scatter plots, and bar charts. Experiment: Create a scatter plot for two numeric variables.

22.	10/04	09/04	09/04	Customizing Graphs in R – Add titles, labels, legends, colors, and themes. Experiment: Customize the plot created in expl6 by adding title, axis labels, and legend.
23.	17/04	16/04	16/04	One-Way ANOVA Using R – Perform and interpret one-way ANOVA. Experiment: Perform one-way ANOVA on a dataset and interpret the result.
24.	17/04	16/04	16/04	Linear Regression Analysis – Implement linear regression, interpret model output, and visualize results. Experiment: Fit a linear regression model and plot the regression line.

Subject Teacher

Principal